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Acknowledgement

I would like to express my sincere gratitude to Metanoia Solutions for providing me with the opportunity to undertake my internship as a Web Developer Intern. This internship has been an invaluable learning experience that allowed me to gain practical exposure to modern web development technologies, software development practices, and the professional work environment.

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During this internship, I successfully designed and developed the Disaster Relief Coordination Platform, a full-stack MERN web application that enables efficient coordination among citizens, volunteers, NGOs, and government authorities during disaster situations. The project enhanced my knowledge of React.js, Node.js, Express.js, MongoDB Atlas, JWT Authentication, Socket.IO, REST APIs, and Agile software development methodology, while strengthening my problem-solving, teamwork, communication, and project management skills.

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Ansh S Patel

ABSTRACT

The Disaster Relief Coordination Platform (DRCP) is a comprehensive, full-stack web application designed to streamline and enhance disaster management operations during emergency situations. This report documents the design, development, implementation, and testing of DRCP, carried out during an internship at Metanoia Solutions from 02 January 2026 to 30 June 2026 under the mentorship of Nehal Patel.

The platform addresses critical gaps in disaster response coordination by providing real-time communication, resource tracking, volunteer management, incident reporting, SOS alerts, and interactive mapping capabilities. Built using the MERN stack (MongoDB, Express.js, React.js, Node.js) with TypeScript, Socket.io for real-time WebSocket communication, and Leaflet.js for GIS mapping, DRCP employs a modular architecture ensuring scalability, security, and ease of deployment. The application features role-based access control for five user roles — Citizens, Volunteers, NGOs, Government Authorities, and Administrators — ensuring secure and appropriate access to sensitive disaster response data.

Key features include multi-language support across 10 Indian languages (including Urdu RTL), progressive web app (PWA) capabilities with offline service worker caching, SOS emergency broadcast system with GPS coordinates, real-time incident tracking, resource allocation and matching, geofencing, CSV bulk import/export, weather integration, interactive dashboards with analytics, and enterprise-grade security features including JWT authentication, Helmet CSP, rate limiting, XSS sanitization, and role-based access control.

Testing was conducted using Vitest for unit and integration tests, along with Playwright for end-to-end testing. Accessibility compliance was verified using @axe-core, achieving WCAG AA standards. The application is deployed on Render with Docker containerization and CI/CD pipelines via GitHub Actions. The platform demonstrates measurable improvements in coordination efficiency and response time compared to conventional manual methods.

LIST OF FIGURES

Figure 1.1 Organization Chart — Metanoia Solutions
Figure 1.2 SDLC Flowchart
Figure 2.1 Department Hierarchy
Figure 2.2 Sequence of Operations Flowchart
Figure 2.3 Production Stages Flowchart
Figure 3.1 Gantt Chart — Internship Schedule
Figure 4.1 Context Level DFD (Level 0)
Figure 4.2 Level 1 DFD
Figure 4.3 Use Case Diagram
Figure 4.4 Activity Diagram — User Registration
Figure 4.5 Activity Diagram — SOS Alert Flow
Figure 4.6 Sequence Diagram — Authentication
Figure 4.7 Sequence Diagram — Request Creation
Figure 5.1 System Architecture Diagram
Figure 5.2 ER Diagram — MongoDB Collections
Figure 5.3 State Transition Diagram
Figure 6.1 Login Page Screenshot
Figure 6.2 Dashboard Screenshot
Figure 6.3 Incident Management Screenshot
Figure 6.4 Interactive Map Screenshot
Figure 6.5 Admin Dashboard Screenshot
Figure 6.6 Resource Management Screenshot
Figure 6.7 SOS Alert Screenshot
Figure 6.8 Profile Page Screenshot

LIST OF TABLES

Table 3.1 Cost Estimation
Table 3.2 Project Dependencies
Table 4.1 Functional Requirements
Table 4.2 Non-Functional Requirements
Table 4.3 Feasibility Study Matrix
Table 4.4 Module Description
Table 5.1 MongoDB Collections and Schema
Table 5.2 API Endpoint Summary
Table 5.3 Security Features Implemented
Table 6.1 Development Environment Specifications
Table 6.2 Technology Specifications
Table 6.3 REST API Implementation Details
Table 7.1 Test Case Document (20 Test Cases)
Table 7.2 Bug Report Summary

Table 8.1 Future Scope Roadmap

TABLE OF CONTENTS

Acknowledgement
Abstract
List of Figures
List of Tables
List of Abbreviations and Symbols
Chapter 1: Overview of the Company
Chapter 2: Company Overview (Detailed)
Chapter 3: Introduction to Internship
Chapter 4: System Analysis
Chapter 5: System Design
Chapter 6: Implementation
Chapter 7: Testing
Chapter 8: Conclusion and Discussion
References
Appendix 1: User Manual
Appendix 2: Installation Guide
Appendix 3: Source Code Structure
Appendix 4: API Documentation
Appendix 5: Database Schema
Daily Work Log
Weekly Progress Report

LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviation	Full Form
API	Application Programming Interface
BCrypt	Blowfish Cryptographic Hash Algorithm
CI/CD	Continuous Integration / Continuous Deployment
CORS	Cross-Origin Resource Sharing
CSP	Content Security Policy
CSRF	Cross-Site Request Forgery
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DOM	Document Object Model
DRCP	Disaster Relief Coordination Platform
ER	Entity Relationship
ERD	Entity Relationship Diagram
GIS	Geographic Information System
GPS	Global Positioning System
GTU	Gujarat Technological University
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IDE	Integrated Development Environment
i18n	Internationalization
IoT	Internet of Things
JSON	JavaScript Object Notation
JWT	JSON Web Token
MERN	MongoDB, Express.js, React.js, Node.js
MongoDB	A NoSQL Document-Oriented Database

NGO	Non-Governmental Organization
NPM	Node Package Manager
ORM	Object-Relational Mapping
PWA	Progressive Web Application
REST	Representational State Transfer
RTL	Right-to-Left (Text Direction)
SDLC	Software Development Life Cycle
SOS	International Distress Signal (Save Our Souls)
SQL	Structured Query Language
SRV	Service Record (DNS)
UAT	User Acceptance Testing
UI/UX	User Interface /

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